



Mobility Agreement

Staff Mobility For Teaching¹

Planned period of the teaching activity: from 26.06.2023 till 30.06.2023

Duration (days) – excluding travel days: 5

The teaching staff member

Last name (s)	Djukanovic	First name (s)	Marko
Seniority ²	senior	Nationality ³	
Gender [Male/Female/Undefined]	Male	Academic year	2022/2023
E-mail	marko.djukanovic@pmf.unibl.org		

The Sending Institution/Enterprise⁴

Name	FH Kärnten		
Erasmus code ⁵ (if applicable)	ASPITTAL01	Faculty/Department	
Address	Europastr. 4; 9524 St. Magdalen, Austria	Country/ Country code ⁶	AT
Contact person name and position	Aleksandra Jama International Relations Office	Contact person e-mail / phone	a.jama@fh-kaernten.at +43 5 90 500 7040
		Size of enterprise (if applicable)	☐ <250 employees ☒ >250 employees

The Receiving Institution

Name	University of Banja Luka	Faculty/Department	Faculty of Sciences
Erasmus code (if applicable)			Mathematics and Informatics
Address	Mladena Stojanovića 2	Country/ Country code	Bosnia and Herzegovina
Contact person name and position	Dragan Matij, mobility staff exchange coordinator	Contact person e-mail / phone	dragan.matic@pmf.unibl.org 0038765 615 945

For guidelines, please look at the end notes on page 3.

Section to be completed BEFORE THE MOBILITY

I. PROPOSED MOBILITY PROGRAMME

Main subject field⁷: 0732

Level (select the main one): Short cycle (EQF level 5) ☐; Bachelor or equivalent first cycle (EQF level 6) ☐;
Master or equivalent second cycle (EQF level 7) ☐; Doctoral or equivalent third cycle (EQF level 8) ☐



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Number of students at the receiving institution benefiting from the teaching programme:

Number of teaching hours⁸:

Language of instruction:

Overall objectives of the mobility:

- 1) Sharing mutual experience
- 2) Professional development of teaching staff
- 3) Providing source of inspiration to others (students and professors)
- 4) Possibility for learning from others

Added value of the mobility (in the context of the modernisation and internationalisation strategies of the institutions involved):

- 1) Strengthening relations between the partner institutions through planning new joint project actions, applying jointly on similar projects, planning new mobility stuff exchanges, as well as new ways to collaborate.

Content of the teaching programme:

Abstract. Mathematical modelling is a non trivial process of expressing business problems in terms of the Algebraic modelling language (AML).

AML is enough expressive to transfer long descriptive problem formulations into general, commonly used mathematical notation via variables, functions, relations between variables, etc.

There are many modelling paradigms known, such as Linear programming, Integer linear programming, Constraint programming, Network programming, etc.

In the context of the teaching programme, we first introduce basic concepts of the most known modelling paradigms through a few real-world examples. Then, some general approaches to solve problems modelled by means of mentioned paradigms will be mentioned. At the end, black-box solvers that directly solve models will be presented and tested on problem instances of various sizes.



Course units:

- 1) Mathematical modelling in general
- 2) Decision variables, constraints, objective functions
- 3) Basics about Linear programming (LP) paradigm
- 4) Solving Linear regression via LP paradigm
- 5) Integer linear programming (ILP) paradigm -- examples
- 6) BnB algorithm for solving ILP
- 7) Genetic algorithm (GA)
- 8) Variable neighbourhood search (VNS) algorithm

Expected outcomes and impact (e.g. on the professional development of the teaching staff member and on the competences of students at both institutions):

For me as a professor:

- 1) Strengthen skills of teaching in an international environment;
- 2) Establish a possible collaboration/contact with (high-motivated) talented students as well as with professors;
- 3) Discuss in live the potential for a research topic and projects that could be fruitful for future collaboration.

For students:

- 1) Students will gain up-to-date information from the science about recent trends in modelling and optimization;
- 2) Students will get insights on how to model and solve problems from theory and practice;
- 3) Students will have a chance to ask in live, thus possibly collaborate with me in the field of optimization and modelling.



II. COMMITMENT OF THE THREE PARTIES

By signing⁹ this document, the teaching staff member, the sending institution/enterprise and the receiving institution confirm that they approve the proposed mobility agreement.

The sending higher education institution supports the staff mobility as part of its modernisation and internationalisation strategy and will recognise it as a component in any evaluation or assessment of the teaching staff member.

The teaching staff member will share his/her experience, in particular its impact on his/her professional development and on the sending higher education institution, as a source of inspiration to others.

The teaching staff member and the beneficiary institution commit to the requirements set out in the grant agreement signed between them.

The teaching staff member and the receiving institution will communicate to the sending institution/enterprise any problems or changes regarding the proposed mobility programme or mobility period.

The teaching staff member

Name: Marko Djukanovic

Signature: *Marko Djukanovic*

Date: 24.02 '23

The sending institution/enterprise

Name of the responsible person: Dragan Matic

Signature: *Dragan Matic*

Date: 24.2.2023.

The receiving institution

Name of the responsible person: Aleksandra Jama

Signature: *A. Jama*

Date: 27.02.2023

Dr. Marko Đukanović
University of Banja Luka
2 Mladena Stojanovića
78000 Banja Luka
Republic of Srpska

Villach, 30 January, 2023

LETTER OF INVITATION

On behalf of the International Relations Office of Carinthia University of Applied Sciences, Austria, I herewith invite

Dr. Marko Đukanović from the Department for Mathematics and Computer Science, Faculty of Natural Science and Mathematics at the University of Banja Luka, Bosnia & Herzegovina,

to Austria.

The visit to Austria is part of an ERASMUS+ project (project number: 2020-1-AT01-KA107-077806) between Carinthia University of Applied Sciences, Austria, and the University of Banja Luka, Bosnia & Herzegovina, implying both student and staff exchange components.

The invitation refers to the time frame of 25 to 30 June 2023.

Costs for accommodation and upkeep will be covered by the project, while health insurance will be covered by Dr. Đukanović himself.

With kind regards,



Mgr. Mgr. Aleksandra Jama MSc
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